# Pro Football Sports Science Research Update

## November 2023

**A narrative review of psychiatric features of traumatic encephalopathy syndrome as conceptualized in the 20th century. Frontiers in Neurology, 14:1214814, 2023.**

**Cannabidiol in sports: Insights on how CBD could improve performance and recovery. Frontiers in Pharmacology, 14:1210202, 2023.**

**Catching a falling star: Mobility of declining star performers, peer effects, and organizational performance in the National Football League. Journal of Business Research, 165:114053, 2023.**

**Determining the true incidence of glenohumeral instability among players in the National Football League – An epidemiological study of non-missed time shoulder instability injuries. The Orthopaedic Journal of Sports Medicine, 11(10): 1-7, 2023.**

**Do non-contact injuries occur during high-speed running in elite football? Preliminary results from a new GPS and video-based method. Journal of Science and Medicine in Sport, Ahead of Print: 1-37, 2023.**

**Does restricted ankle joint mobility influence hamstring muscle strength, work and power in football players after ACL reconstruction and non-injured players? Journal of Clinical Medicine, 12:6330, 2023.**

**Effects of COVID-19 on rate of injury and position-specific injury during the 2020 National Football League season. The Journal of Strength and Conditioning Research, Ahead of Print: 1-8, 2023.**

**Exploring the factors involved in being "Ready" to return to sport following a concussion. Clinical Journal of Sport Medicine, 33:395-404, 2023.**

**Hamstring strain injury risk factors in Australian football change over the course of the season. Medicine and Science in Sports and Exercise, Ahead of Print: 1-45, 2023.**

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**MRI radiomics for hamstring strain injury identification and return to sport classification: A pilot study. Skeletal Radiology, Ahead of Print: 1-12, 2023.**

**Narrative review: The role of circadian rhythm on sports performance, hormonal regulation, immune system function, and injury prevention in athletes. Heliyon, 9:e19636, 2023.**

**Percentile rankings and position differences for absolute and allometrically scaled performance measures from the National Football League Scouting Combine. The Journal of Strength and Conditioning Research, Ahead of Print: 1-12, 2023.**

**Performance and symmetry measures during vertical jump testing at return to sport after ACL reconstruction. British Journal of Sports Medicine, Ahead of Print: 1-8, 2023.**

**Popular dietary trends' impact on athletic performance: A critical analysis review. Nutrients, 15: 3511, 2023.**

**Preliminary brain-behavioral neural correlates of anterior cruciate ligament injury risk landing biomechanics using a novel bilateral leg press neuroimaging paradigm. PLOS ONE, 17(8): e0272578, 2023.**

**Prevalence of adulteration in dietary supplements and recommendations for safe supplement practices in sport. Frontiers in Sports and Active Living, 5: 1239121, 2023.**

**Sports Hernias: A comprehensive review for clinicians. Cureus, 15(8): e43283, 2023.**

**The impact of sleep interventions on athletic performance: A systematic review. Sport Medicine – Open, 9:58, 2023.**

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# Pro Football Sports Science Research Update

## *A narrative review of psychiatric features of traumatic encephalopathy syndrome as conceptualized in the 20th century. Frontiers in Neurology, 14:1214814, 2023.*

**Introduction:** Some ultra-high exposure boxers from the 20th century suffered from neurological problems characterized by slurred speech, personality changes (e.g., childishness or aggressiveness), and frank gait and coordination problems, with some noted to have progressive Parkinsonian-like signs. Varying degrees of cognitive impairment were also described, with some experiencing moderate to severe dementia. The onset of the neurological problems often began while they were young men and still actively fighting. More recently, traumatic encephalopathy syndrome (TES) has been proposed to be present in athletes who have a history of contact (e.g., soccer) and collision sport participation (e.g., American-style football). The characterization of TES has incorporated a much broader description than the neurological problems described in boxers from the 20th century. Some have considered TES to include depression, suicidality, anxiety, and substance abuse.

**Purpose:** We carefully re-examined the published clinical literature of boxing cases from the 20th century to determine whether there is evidence to support conceptualizing psychiatric problems as being diagnostic clinical features of TES.

**Methods:** We reviewed clinical descriptions from 155 current and former boxers described in 21 articles published between 1928 and 1999.

**Results:** More than one third of cases (34.8%) had a psychiatric, neuropsychiatric, or neurobehavioral problem described in their case histories. However, only 6.5% of the cases were described as primarily psychiatric or neuropsychiatric in nature. The percentages documented as having specific psychiatric problems were as follows: depression = 11.0%, suicidality = 0.6%, anxiety = 3.9%, anger control problems = 20.0%, paranoia/suspiciousness = 11.6%, and personality change = 25.2%.

**Discussion:** We conclude that depression, suicidality (i.e., suicidal ideation, intent, or planning), and anxiety were not considered to be clinical features of TES during the 20th century. The present review supports the decision of the consensus group to remove mood and anxiety disorders, and suicidality, from the new 2021 consensus core diagnostic criteria for TES. More research is needed to determine if anger dyscontrol is a core feature of TES with a clear clinicopathological association. The present findings, combined with a recently published large clinicopathological association study, suggest that

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mood and anxiety disorders are not characteristic of TES and they are not associated with chronic traumatic encephalopathy neuropathologic change.

### *Cannabidiol in sports: Insights on how CBD could improve performance and recovery. Frontiers in Pharmacology, 14:1210202, 2023.*

**Overview/Conclusions -** CBD appears to have anti-inflammatory, neuroprotective, analgesic, anxiolytic, and potentially recovery-mediating properties in athletes, but more scientific evidence is needed to confirm these effects. Confirmatory analyses using randomized controlled trials with placebo are necessary to test the acute and chronic effects of different dosage prescriptions. These studies must consider fundamental sport-specific particularities, such as the diverse biological and situational conditions that contribute to fatigue, the characteristics of each discipline during training and competition, the individual peculiarities of athletes, their tolerance and response to CBD intake, and the combined effect of CBD administration with other physical and nutritional aids.

Given the relatively common use of cannabis and CBD among athletes, there is a clear need to improve the scientific understanding of the effects of CBD use on athlete recovery and performance. Further scientific progress is necessary, primarily through the execution of experimental trials, to better understand critical positive and negative outcomes for the ultimate benefit of athlete recovery and performance. Furthermore, resulting evidence could provide new clinical guidance for prescribing CBD during the athlete recovery process and other potential applications. The potential therapeutic benefits of CBD administration have been minimized for years, but the actual scenario could increase knowledge about this natural compound and its effects. Additionally, from an administrative point of view, adopting a clearer and more global policy for the use of cannabis in sports should be considered.

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## *Catching a falling star: Mobility of declining star performers, peer effects, and organizational performance in the National Football League. Journal of Business Research, 165:114053, 2023.*

**Abstract -** Research examining how star performers impact their peers largely finds that stars positively impact peer performance. Another stream of research has examined how stars impact their organizations, specifically after mobility events, and finds that hiring stars often leads to lower organizational performance. However, these research streams have developed independently. This paper attempts to reconcile these findings by examining the mobility of stars in the twilight of their career. Leveraging a dataset of individual and organizational performance in the NFL, we find that falling star quarterbacks have positive impacts on their peers when they change organizations. This peer effect leads to increased organizational performance, even though the individual performance of the falling star declines. Moreover, we test and compare the specific mechanisms driving this effect.

## *Determining the true incidence of glenohumeral instability among players in the National Football League – An epidemiological study of non-missed time shoulder instability injuries. The Orthopaedic Journal of Sports Medicine, 11(10): 1-7, 2023.*

**Background:** Shoulder instability encompasses a spectrum of glenohumeral pathology ranging from subluxation to dislocation. While dislocation frequently leads to removal from play, athletes are often able to play through subluxation. Previous research on glenohumeral instability among athletes has largely focused on missed-time injuries, which has likely disproportionately excluded subluxation injuries and underestimated the overall incidence of shoulder instability.

**Purpose:** To describe the epidemiology of shoulder instability injuries resulting in no missed time beyond the date of injury (non– missed time injuries) among athletes in the National Football League (NFL).

**Study Design:** Descriptive epidemiology study.

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**Methods:** The NFL's electronic medical record was retrospectively reviewed to identify non–missed time shoulder instability injuries during the 2015 through 2019 seasons. For each injury, player age, player position, shoulder laterality, instability type, instability direction, injury timing, injury setting, and injury mechanism were recorded. For injuries that occurred during games, incidence rates were calculated based on time during the season as well as player position. The influence of player position on instability direction was also investigated.

**Results:** Of the 546 shoulder instability injuries documented during the study period, 162 were non–missed time injuries. The majority of non–missed time injuries were subluxations (97.4%), occurred during games (70.7%), and resulted from a contact mechanism (91.2%). The overall incidence rate of game-related instability was 1.6 injuries per 100,000 player-plays and was highest during the postseason (3.5 per 100,000 player-plays). The greatest proportion of non–missed time injuries occurred in defensive secondary players (28.4%) and offensive linemen (19.8%), while kickers/punters and defensive secondary players had the highest game incidence rates (5.5 and 2.1 per 100,000 player-plays, respectively). In terms of direction, 54.3% of instability events were posterior, 31.9% anterior, 8.5% multidirectional, and 5.3% inferior. Instability events were most often anterior among linebackers and wide receivers (50% and 100%, respectively), while posterior instability was most common in defensive linemen (66.7%), defensive secondary players (58.6%), quarterbacks (100.0%), running backs (55.6%), and tight ends (75.0%).

**Conclusion:** The majority of non–missed time shoulder instability injuries (97.4%) were subluxations, which were likely excluded from or underreported in previous shoulder instability studies due to the inherent difficulty of detecting and diagnosing shoulder subluxation.

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## *Do non-contact injuries occur during high-speed running in elite football? Preliminary results from a new GPS and video-based method. Journal of Science and Medicine in Sport, Ahead of Print: 1-37, 2023.*

**Objectives:** Understanding how injuries occur (inciting circumstances) is useful for developing etiological hypotheses and prevention strategies. The aims of this study were 1) to evaluate the feasibility of a method combining video and Global Positioning System data to estimate the speed and acceleration of activities leading to injuries and 2) to use this method to analyze the inciting circumstances leading to non-contact injuries.

**Design:** Retrospective descriptive study.

**Methods:** Injury inciting circumstances from 46 elite players over three seasons were analyzed from video recordings and from external load measures collected through Catapult Vector S7 Global Positioning System.

**Results:** In total 34 non-contact injuries were analyzed. Sixteen out of the seventeen hamstring injuries occurred when players were running for (median and interquartile range) 16.75 m (8.42-26.65 m) and achieved a peak speed of 29.28 km·h-1 (26.61-31.13 km·h-1) which corresponded to 87.55 % of players' maximal speed (78.5 %-89.75 %). Of the three adductor injuries, one occurred whilst the player was decelerating without the ball, one occurred whilst the player was accelerating and controlling the ball at knee level, and one occurred whilst the player was performing an instep kick. Two quadriceps injuries occurred whilst the players were kicking either whilst walking or running.

**Conclusions:** From the preliminary results reported in this study most hamstring injuries occurred when players ran >25 km·h-1 and above 80 % of their maximal speed. This study suggests that this novel approach can allow a detailed and standardized analysis of injury inciting circumstances.

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## *Does restricted ankle joint mobility influence hamstring muscle strength, work and power in football players after ACL reconstruction and non-injured players? Journal of Clinical Medicine, 12:6330, 2023.*

**Abstract -** This study was aimed at observing how the limitation of ankle dorsiflexion ROM affects hamstring muscle Peak Torque/BW (%), Average Power (W), and Total Work (J), and whether this effect is similar in football players after ACL rupture and reconstruction and in those without injuries. The study included 47 professional football players who were divided into two groups: Group 1 (n = 24) after ACL reconstruction and Group 2 (n = 23) without injuries in the past 3 years. Based on the Weight-Bearing Lunge Test (WBLT), the following subgroups in Groups 1 and 2 were distinguished: N (normal ankle joint dorsiflexion) and R (restricted ankle joint dorsiflexion). The concentric isokinetic test (10 knee flexions and extensions at 60◦/s) was performed on both limbs. Significantly lower values of Peak Torque/BW and Average Power were observed in Group 1 compared to Group 2, as well as in subjects with normal and restricted ankle dorsiflexion. However, no significant differences were noted for either group in any of the strength variables comparing subjects with normal and restricted ankle dorsiflexion. A poor and non-significant correlation was exhibited between the ankle joint range of dorsiflexion and all the strength variables. The area under the ROC curve (AUC) for all the evaluated variables in both groups was below 0.5, or very close to this value, indicating that ankle dorsiflexion ROM has no diagnostic accuracy for hamstring muscle strength. Based on the obtained results, it can be assumed that ankle dorsiflexion limitation, which is common in football players, is not a factor in weakening hamstring muscle strength, either in football players after ACL reconstruction or among those without injuries. However, some authors have reported that limited mobility of the ankle joint can have a destructive effect on the work of the lower limbs and may also be a factor in increasing the risk of football injuries in this area. Therefore, we have suggested that hamstring muscle weakness and increased risk of injury may occur due to factors other than limited ankle mobility. These observations may be of great importance in the selection of prevention methods by including a broad spectrum of physical techniques, not just exercises that focus on the improvement of mobility or stability of the lower limbs.

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## *Effects of COVID-19 on rate of injury and position-specific injury during the 2020 National Football League season. The Journal of Strength and Conditioning Research, Ahead of Print: 1-8, 2023.*

**Abstract -** Because of the COVID-19 pandemic, the National Football League (NFL) made changes to its operations for the 2020 season. We hypothesize an increase in the rate of injuries during the 2020 season. Publicly available data were reviewed to identify NFL injuries from the 2015–2020 seasons. Player position, description of injury, date of injury, and injury setting were recorded. $p \leq 0.05$ was considered statistically significant. For the 2020 season, compared with the 2015–2019 seasons, there was an increased risk of injury during the regular season overall relative risk (RR) 5 1.308 ($p < 0.05$), week (W)1 RR 5 7.33 ($p < 0.05$), W1–6 RR 5 1.964 ($p < 0.05$), W7–12 RR 5 1.8909 (p , 0.05), and during the postseason overall RR 5 1.1444 ($p < 0.05$), calculated using analysis of variance. There was an overall increased risk of abdominal or core injuries RR 5 1.248 ($p < 0.05$), groin or hip injuries RR 5 2.534 ($p < 0.05$), and hamstring injuries RR 5 3.644 ($p < 0.05$). There was an increased risk of hamstring injuries in cornerbacks RR 5 3.219 ($p < 0.05$) and running backs RR 5 1.1394 ($p < 0.05$), hip or groin injuries in guards RR 5 1.105 ($p < 0.05$), Achilles tendon injuries in safeties RR 5 1.6976 ($p < 0.05$), quadriceps injuries in running backs RR 5 1.6191 ($p < 0.05$), and arm injuries in defensive tackles RR 5 1.221 ($p < 0.05$). There was an increase in the overall rate of injuries in the 2020 NFL season, both in the regular season and postseason, compared with the 2015–2019 seasons. The overall rate of abdominal or core, groin or hip, and hamstring injuries increased. Specific player positions saw unique increases in rates of injuries. These findings may be due to numerous operational changes implemented, such as reduced in-person training and the elimination of the preseason, leading to suboptimal, sports-specific conditioning and increased risk of musculoskeletal injury.

**Practical Applications -** Although the NFL adapted their operations during the COVID 19 pandemic, the changes made to combat the spread of infection may have contributed to the increase in injuries during the 2020 NFL season. Ultimately, alterations to players' traditional offseason training regimens may have led to suboptimal conditioning and preparation for the season. An increased rate of injuries was observed both overall for the regular season, with a notable increase early in the regular season, and overall, for the postseason, when compared with previous NFL seasons. Guards and cornerbacks experienced an increase in overall positional injuries, whereas there were also several specific anatomic injuries in unique position groups which increased. These increases in position-specific anatomic injuries included hamstring injuries in cornerbacks and running backs, Achilles injuries in safeties, quadriceps

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injuries in running backs, hip or groin injuries in guards, and arm injuries in defensive tackles. There was no observed change in the proportion of overall injuries observed in games versus practices. The results of this study highlight the need for proper conditioning and a gradual increase in physical load for athletes across sports. Coaches and athletes alike should be aware of these findings to guide training regimens, practices, and game schedules. Caution should be taken when modifying traditional training schedules, including switching to virtual workouts. These findings are particularly relevant because with the increase in virtual communication and training platforms, individuals and teams will undoubtably attempt to leverage this technology, increasing the time spent away from team facilities during the offseason. This study can additionally be used to understand the epidemiology of NFL injuries by health care professionals. These results enable them to customize treatment protocols that allow for the maximization of playing time and performance for players while reducing risk of injury. Finally, athletic trainers, physical therapists, and coaches can use this information to identify particularly high-risk player positions, as well as their most common anatomic injuries, allowing the creation of optimized workout plans to reduce incidence of specific injuries.

## *Exploring the factors involved in being "Ready" to return to sport following a concussion. Clinical Journal of Sport Medicine, 33:395-404, 2023.*

**Objective:** To explore the factors involved in athletes being ready (or not) to return to sport (RTS) after sport-related concussion (SRC).

**Design:** Qualitative, semi-structured interviews.

**Setting:** Videoconference.

**Participants:** Twenty-two sport-injury stakeholders involved in contact and collision sports at various levels of competition (high school, university, professional), including: formerly concussed athletes (n = 4), coaches (n = 5), athletic therapists (n = 5), physiotherapists (n = 4), nurse practitioner (n = 1), and sports medicine physicians (n = 3).

**Interventions:** N/A.

**Main Outcome Measures:** We included questions in the interview guide regarding factors participants believed were involved in athletes being ready (or not ready) to RTS after a concussion.

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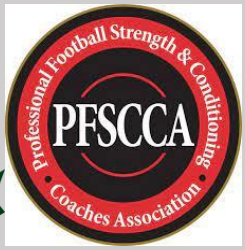

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**Results:** Participants described physical (concussion symptoms, return to pre-injury fitness), behavioral (changes in behavior, avoidance, malingering), psychological (individual factors, cognitive appraisals, mental health), and social (isolation, social support, communication, pressure) factors that they believed were involved in athletes being ready to RTS after SRC.

**Conclusions:** The graduated RTS strategy outlined in the most recent Concussion in Sport Group consensus statement focuses on physical aspects involved in being ready to RTS, which does not address behavioral, psychological, and social factors, which were identified by participants as being related to returning to sport post-SRC. More research is needed to determine whether the additional factors outlined in this study are relevant among larger samples of athletes, coaches, and healthcare professionals.

## *Hamstring strain injury risk factors in Australian football change over the course of the season. Medicine and Science in Sports and Exercise, Ahead of Print: 1-45, 2023.*

**Background/aim:** To determine which factors were most predictive of hamstring strain injury (HSI) during different stages of the competition in professional Australian Football.

**Methods:** Across two competitive seasons, eccentric knee flexor strength and biceps femoris long head (BFlh) architecture of 311 Australian Football players (455 player seasons) were assessed at the start and end of pre-season and in the middle of the competitive season. Details of any prospective HSIs were collated by medical staff of participating teams. Multiple logistic regression models were built to identify important risk factors for HSI at the different time points across the season.

**Results:** There were 16, 33 and 21 new HSIs reported in preseason, early in-season, and late in-season, respectively across two competitive seasons. Multivariate logistic regression and recursive feature selection revealed that risk factors were different for pre-season, early in-season and late in-season HSIs. A combination of prior HSI, age, height and muscle thickness were most associated with pre-season injuries (median AUC, 0.83). Pennation angle and fascicle length had the strongest association with early in-season injuries (median AUC, 0.86). None of the input variables were associated with late in-season injuries (median AUC, 0.46). Identification of early in-season HSIs and late in-season HSIs was not improved by the magnitude of change of data across pre-season (median AUC, 0.67).

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**Conclusions:** Risk factors associated with prospective HSIs were different across the season in Australian Rules Football, with non-modifiable factors (prior HSI, age, and height) mostly associated with pre-season injuries. Early in-season HSIs were associated with modifiable factors, notably BFlh architectural. measures. The prediction of in-season HSIs was not improved by assessing the magnitude of change in data across pre-season

## *Interaction of surface type, temperature, and week of season on concussion risk in the National Football League. Epidemiology, 34: 807-816, 2023.*

**Background:** Artificial turf fields and environmental conditions may influence sports concussion risk, but existing research is limited by uncontrolled confounding factors, limited sample size, and the assumption that risk factors are independent of one another. The purpose of this study was to examine how playing surface, time of season, and game temperature relate to diagnosed concussion risk in the National Football League (NFL).

**Methods:** This retrospective cohort study examined data from the 2012 to the 2019 NFL regular season. We fit Bayesian negative binomial regression models to relate how playing surface, game temperature, and week of the season independently related to diagnosed concussion risk and any interactions among these factors.

**Results:** We identified 1096 diagnosed concussions in 1830 games. There was a >99% probability that concussion risk was reduced on grass surface (median incidence rate ratio [IRR] = 0.78 [95% credible interval: 0.68, 0.89]), >99% probability that concussion risk was lower at higher temperatures (IRR = 0.85 [0.76,0.95] for each 7.9 °C), and >91% probability that concussion risk increased with each week of the season (IRR = 1.02 [1.00,1.04]). There was an >84% probability for a surface × temperature interaction (IRR = 1.01 [0.96, 1.28]) and >75% probability for a surface × week interaction (IRR = 1.02 [0.99, 1.05]).

**Conclusions:** Diagnosed concussion risk is increased on artificial turf compared with natural grass, and this is exacerbated in cold weather and, independently, later in the season. The complex interplay between these factors necessitates accounting for multiple factors and their interactions when investigating sports injury risk factors and devising mitigation methods.

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# Pro Football Sports Science Research Update

## *Match running performance in Australian football is related to muscle fiber typology. International Journal of Sports Physiology and Performance, Ahead of Print: 1-7, 2023.*

**Purpose:** To examine the association between muscle fiber typology and match running performance in professional Australian football (AF) athletes.

**Methods:** An observational time–motion analysis was performed on 23 professional AF athletes during 224 games throughout the 2020 competitive season. Athletes were categorized by position as hybrid, small, or tall. Athlete running performance was measured using Global Navigation Satellite System devices. Mean total match running performance and maximal mean intensity values were calculated for moving mean durations between 1 and 10 minutes for speed (in meters per minute), high-speed-running distance (HSR, >4.17 $m \cdot s^{-1}$ ), and acceleration (in meters per second squared), while intercept and slopes were calculated using power law. Carnosine content was quantified by proton magnetic resonance spectroscopy in the gastrocnemius and soleus and expressed as a carnosine aggregate z score (CAZ score) to estimate muscle fiber typology. Mixed linear models were used to determine the association between CAZ score and running performance.

**Results:** The mean (range) CAZ score was −0.60 (−1.89 to 1.25), indicating that most athletes possessed a greater estimated proportion of type I muscle fibers. A greater estimated proportion of type I fibers (ie, higher CAZ score) was associated with a larger accumulation of HSR (>4.17 $m \cdot s^{-1}$ ) and an increased ability to maintain HSR as the peak period duration increased.

**Conclusion:** AF athletes with a greater estimated proportion of type I muscle fibers were associated with a greater capacity to accumulate distance running at high speeds, as well as a greater capacity to maintain higher output of HSR running during peak periods as duration increases.

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## *MRI radiomics for hamstring strain injury identification and return to sport classification: A pilot study. Skeletal Radiology, Ahead of Print: 1-12, 2023.*

**Objective:** To determine if MRI-based radiomics from hamstring muscles are related to injury and if the features could be used to perform a time to return to sport (RTS) classification. We hypothesize that radiomics from hamstring muscles, especially T2-weighted and diffusion tensor imaging-based features, are related to injury and can be used for RTS classification.

**Subjects and methods:** MRI data from 32 athletes at the University of Wisconsin-Madison that sustained a hamstring strain injury were collected. Diffusion tensor imaging and T1- and T2-weighted images were processed, and diffusion maps were calculated. Radiomics features were extracted from the four hamstring muscles in each limb and for each MRI modality, individually. Feature selection was performed and multiple support vector classifiers were cross-validated to differentiate between involved and uninvolved limbs and perform binary (≤ or>25 days) and multiclass (42 days) classification of RTS.

**Result:** The combination of radiomics features from all diffusion tensor imaging and T2-weighted images provided the most accurate differentiation between involved and uninvolved limbs (AUC ≈ 0.84±0.16). For the binary RTS classification, the combination of all extracted radiomics offered the most accurate classification (AUC ≈ 0.95±0.15). While for the multiclass RTS classification, the combination of features from all the diffusion tensor imaging maps provided the most accurate classification (weighted one vs. rest AUC ≈ 0.81±0.16).

**Conclusion:** This pilot study demonstrated that radiomics features from hamstring muscles are related to injury and have the potential to predict RTS.

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## *Narrative review: The role of circadian rhythm on sports performance, hormonal regulation, immune system function, and injury prevention in athletes. Heliyon, 9:e19636, 2023.*

**Objectives:** This study was a narrative review of the importance of circadian rhythm (CR), describes the underlying mechanisms of CR in sports performance, emphasizes the reciprocal link between CR, endocrine homeostasis and sex differences, and the unique role of the circadian clock in immune system function and coordination.

**Method:** As a narrative review study, a comprehensive search was conducted in PubMed, Scopus, and Web of Science (core collection) databases using the keywords “circadian rhythm”, “sports performance”, “hormonal regulation”, “immune system”, and “injury prevention”. Inclusion criteria were studies published in English and peer-reviewed journals until July 2023. Studies that examined the role of CR in sports performance, hormonal status, immune system function, and injury prevention in athletes were selected for review.

**Results:** CR is followed by almost all physiological and biochemical activities in the human body. In humans, the superchiasmatic nucleus controls many daily biorhythms under solar time, including the sleep-wake cycle. A body of literature indicates that the peak performance of essential indicators of sports performance is primarily in the afternoon hours, and the evening of actions occurs roughly at the peak of core body temperature. Recent studies have demonstrated that the time of day that exercise is performed affects the achievement of good physical performance. This review also shows various biomarkers of cellular damage in weariness and the underlying mechanisms of diurnal fluctuations. According to the clock, CR can be synchronized with photonic and non-photonic stimuli (i.e., temperature, physical activity, and food intake), and feeding patterns and diet changes can affect CR and redox markers. It also emphasizes the reciprocal links between CR and endocrine homeostasis, the specific role of the circadian clock in coordinating immune system function, and the relationship between circadian clocks and sex differences.

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# Pro Football Sports Science Research Update

## *Percentile rankings and position differences for absolute and allometrically scaled performance measures from the National Football League Scouting Combine. The Journal of Strength and Conditioning Research, Ahead of Print: 1-12, 2023.*

**Abstract -** This study aimed to provide percentile rankings for absolute and allometrically scaled performance measures for National Football League (NFL) Scouting Combine subjects and determine if allometric scaling affected position-specific differences in performance measures. Subjects included American football players (n = 3,015) who participated in the NFL Scouting Combines between 2015 and 2019. Subjects were divided into position groups: defensive backs (DBs, n = 562), defensive linemen (DL, n = 498), linebackers (LBs, n = 395), offensive linemen (OL, n = 505), running backs (RBs, n = 303), tight ends (TEs, n = 165), quarterbacks (QBs, n = 165), and wide receivers (WRs, n = 422). Performance measures included 40-yd dash time (with 10- and 20-yd split times), bench press repetitions to failure, vertical jump height, broad jump distance, pro-agility time, and L-cone drill time. Descriptive statistics were computed for all performance metrics and allometrically scaled performance metrics. One-way analyses of variance tested for position group differences for absolute and allometrically scaled performance measures. Percentile rankings for absolute and allometrically scaled performance measures were determined. Position-specific differences for absolute performance measures indicated that DB and WR tended to be faster, jump higher and further, and have greater change-of-direction capabilities compared with LB, QB, RB, and TE, who scored better in these respects than OL and DL. Allometric scaling revealed that DL, LB, RB, and TE tended to have superior performance than DB and WR, with QB and OL having the poorest performance. This study provides percentile rankings and the equation and parameters by which coaches and practitioners may allometrically scale data for evaluations of normalized performance measures from the NFL Scouting Combine.

**Practical Applications -** Coaches and practitioners who seek to determine the performance level of NFL Scouting Combine subjects should use normative reference values (percentile rankings) that allow comparisons to recent NFL Scouting Combine subjects. Furthermore, many coaches and practitioners seek not only to examine absolute performance measures but also to normalize performance to body mass to allow comparisons of players of different positions and performance levels. For example, if a WR jumps higher than a DL, yet the DL has greater normalized athletic performance, it becomes evident that the only reason differences in jump height exist because of the confounding effect of body mass. This suggests that although the WR jumped higher, the DL likely has greater power output. It is possible

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that this discrepancy in normalized jump height, and potentially power output, may be the result of differences in underlying muscle mass, providing greater information regarding position-related differences not found by assessments of absolute performance. These types of comparisons of normalized performance measures may help guide strength and conditioning professionals regarding position-specific strength and conditioning programs for American football players. This study provides the necessary equation and allometric parameters by which coaches and practitioners may normalize NFL Scouting Combine data effectively for future evaluations of normalized athletic performance among elite American football players.

### *Performance and symmetry measures during vertical jump testing at return to sport after ACL reconstruction. British Journal of Sports Medicine, Ahead of Print: 1-8, 2023.*

**Objective:** Vertical jump tests are more sensitive in revealing asymmetries in performance metrics at the time of return to sport after anterior cruciate ligament (ACL) reconstruction (ACLR) than horizontal hop tests. However, it remains unclear which vertical tests (bilateral or unilateral) and which metrics (kinetics or performance) are most effective in informing the rehabilitation status and readiness for return to sport. We aimed to investigate the status of athletes during vertical jump testing at return to sport after ACLR.

**Methods:** A dual force platform system was used to evaluate jumping performance of 126 recreational and professional athletes at the time of return to sport after ACLR, as well as 532 healthy control participants. Performance and kinetic metrics were collected during four jump tests: double-leg countermovement jump, single-leg countermovement jump, double-leg 30 cm drop jump and single-leg 15 cm drop jump. Between limb and between-group differences were explored using mixed models analyses.

**Results:** At the time of return to sport after ACLR, athletes still presented significant differences favoring the uninvolved side, particularly in the symmetry of the concentric impulse ($p<0.001$) in all jumps compared with the control group. Peak landing force asymmetry was greater in the ACLR group than the controls during the countermovement ($p<0.001$, MD=−11.6; 95% CI –15.4 to –7.9) and the double-leg drop jump ($p=0.023$, MD=−8.9; 95% CI –14.9 to –2.8). The eccentric impulse asymmetry was significantly

greater (p=0.018, MD=−3.8; 95% CI −5.8 to –1.7) in the ACLR group during the single-leg drop jump only. Jump height was significantly lower (p<0001) in the ACLR group compared with controls in all tests except the double-leg drop jump.

**Conclusion:** At the time of return to sport after ACLR, despite passing the traditional discharge criteria, athletes remained asymmetrical during all vertical jump tests, in the concentric (push-off) phase, during landing from bilateral jumps and for most performance metrics. Clinicians should aim to restore not only symmetry in ground reaction forces but also absolute performance metrics such as jump height, reactive strength index and contact times, to potentially reduce injury risk and improve overall athletic performance.

**WHAT IS ALREADY KNOWN ON THIS TOPIC**

⇒ At the time of return to sport, athletes still present biomechanical asymmetries (joint angles, moments and work and muscle forces) during vertical jumps, despite passing discharge criteria.

⇒ Ground reaction force analysis during vertical jump testing can offer valuable phase-specific information on the status of an athlete after anterior cruciate ligament reconstruction (ACLR).

**WHAT THIS STUDY ADDS**

⇒ We report the metrics that are not yet restored at the time of return to sport, and therefore, likely to be of clinical interest.

⇒ Asymmetries were present in the concentric phase (push-off), during landing and in most of the performance metrics.

⇒ We provide normative data of performance metrics during vertical jumps of a large cohort of professional football players.

**HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY**

⇒ Vertical jumps (single and bilateral) should be included in the periodic and return to sport testing after ACLR.

⇒ Clinicians should monitor and restore symmetry and absolute performance metrics such as jump height, reactive strength index and contact times.

# Pro Football Sports Science Research Update

⇒ Plyometric training should be an integral part of the rehabilitation protocol given the observed residual deficits seen in the single-leg drop jump test.

## *Popular dietary trends' impact on athletic performance: A critical analysis review. Nutrients, 15: 3511, 2023.*

**Background:** Nutrition fuels optimal performance for athletes. With increased research developments, numerous diets available, and publicity from professional athletes, a review of dietary patterns impact on athletic performance is warranted.

**Results:** The Mediterranean diet is a low inflammatory diet linked to improved power and muscle endurance and body composition. Ketogenic diets are restrictive of carbohydrates and proteins. Though both show no decrements in weight loss, ketogenic diets, which is a more restrictive form of low-carbohydrate diets, can be more difficult to follow. High-protein and protein-paced versions of low-carbohydrate diets have also shown to benefit athletic performance. Plant-based diets have many variations. Vegans are at risk of micronutrient deficiencies and decreased leucine content, and therefore, decreased muscle protein synthesis. However, the literature has not shown decreases in performance compared to omnivores. Intermittent fasting has many different versions, which may not suit those with comorbidities or specific needs as well as lead to decreases in sprint speed and worsening time to exhaustion.

**Conclusions:** This paper critically evaluates the research on diets in relation to athletic performance and details some of the potential risks that should be monitored. No one diet is universally recommend for athletes; however, this article provides the information for athletes to analyze, in conjunction with medical professional counsel, their own diet and consider sustainable changes that can help achieve performance and body habitus goals.

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## *Preliminary brain-behavioral neural correlates of anterior cruciate ligament injury risk landing biomechanics using a novel bilateral leg press neuroimaging paradigm. PLOS ONE, 17(8): e0272578, 2023.*

**Abstract -** Anterior cruciate ligament (ACL) injury risk reduction strategies primarily focus on biomechanical factors related to frontal plane knee motion and loading. Although central nervous system processing has emerged as a contributor to injury risk, brain activity associated with the resultant ACL injury-risk biomechanics is limited. Thus, the purposes of this preliminary study were to determine the relationship between bilateral motor control brain activity and injury risk biomechanics and isolate differences in brain activity for those who demonstrate high versus low ACL injury risk. Thirty-one high school female athletes completed a novel, multi-joint leg press during brain functional magnetic resonance imaging (fMRI) to characterize bilateral motor control brain activity. Athletes also completed an established biomechanical assessment of ACL injury risk biomechanics within a 3D motion analysis lab oratory. Knee abduction moments during landing were modelled as a covariate of interest within the fMRI analyses to identify directional relationships with brain activity and an injury risk group classification analysis, based on established knee abduction moment cut-points. Greater landing knee abduction moments were associated with greater lingual gyrus, intracalcarine cortex, posterior cingulate cortex and precuneus activity when performing the bilateral leg press (all $z > 3.1$, $p < .05$; multiple comparison corrected). In the follow-up injury-risk classification analysis, those classified as high ACL injury-risk had greater activity in the lingual gyrus, parietal cortex and bilateral primary and secondary motor cortices relative to those classified as low ACL injury-risk (all $z > 3.1$, $p < .05$; multiple comparison corrected). In young female athletes, elevated brain activity for bilateral leg motor control in regions that integrate sensory, spatial, and attentional information were related to ACL injury-risk landing biomechanics. These data implicate crossmodal visual and proprioceptive integration brain activity and knee spatial awareness as potential neurotherapeutic targets to optimize ACL injury-risk reduction strategies.

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# Pro Football Sports Science Research Update

## *Prevalence of adulteration in dietary supplements and recommendations for safe supplement practices in sport. Frontiers in Sports and Active Living, 5: 1239121, 2023.*

**Abstract -** The prevalence of dietary supplement use among athletes continues to rise with 60–80% of athletes often reporting current or previous use of dietary supplements. While select dietary ingredients have been shown to improve acute performance and enhance training adaptations over time, it is important to still consider the risk vs. reward for athletes before opting to consume a dietary supplement. Previous work has indicated that certain dietary supplements may pose risks for inadvertent doping, may be susceptible to mislabeling, could be banned by certain governing bodies of sport, or pose health risks for certain populations. The purpose of the current narrative review is to summarize the prevalence of adulteration in dietary sport supplement products, outline the risks of inadvertent doping for athletes, and highlight best practices regarding safe supplementation strategies. Analytical studies have found anywhere from 14 to 50% of samples analyzed from dietary supplement products have tested positive for anabolic agents or other prohibited substances. It is important for the consumer to adhere to safe supplementation strategies, which include following serving size recommendations, cross-referencing ingredient profiles with the list of prohibited substances, choosing quality products that have been verified by a third-party certification program, and being cognizant of consuming multiple dietary supplement products with overlapping ingredient profiles. Once these practices have been considered, it is reasonable for an athlete to utilize dietary supplements as a strategy to optimize performance and health, with a low risk of failing a drug test (adverse analytical finding) and experiencing adverse events.

## *Sports Hernias: A comprehensive review for clinicians. Cureus, 15(8): e43283, 2023.*

**Overview -** Sports hernias represent a complex and often under-recognized cause of chronic groin pain in athletes, which can significantly impair their performance and quality of life. Despite the myriad of terms used to describe this condition, the underlying pathophysiology of sports hernias remains elusive, as no single anatomical structure is consistently implicated in its development. This lack of consensus has led to diagnostic challenges and varying treatment approaches, making the management of sports

hernias a topic of ongoing debate among clinicians. In recent years, there has been an increased interest in understanding the etiology, risk factors, and biomechanics of sports hernias to improve diagnostic accuracy and tailor treatment strategies. The growing body of research has elucidated the multifactorial nature of this condition, implicating a combination of muscle imbalances, core instability, and pelvic dysfunction in the genesis of sports hernias. Moreover, the emerging role of advanced imaging modalities has shed light on the subtle anatomical changes that characterize these injuries, enabling clinicians to identify affected structures with greater precision. The purpose of this comprehensive review is to provide clinicians with an updated understanding of sports hernias in the context of the latest advancements in the field. We will begin by delving into the current theories on pathogenesis, highlighting the diverse factors that contribute to the development of sports hernias. Next, we will explore the diagnostic process, focusing on the role of clinical examination, imaging studies, and the use of diagnostic algorithms to achieve a more accurate and timely diagnosis. Finally, we will discuss the management of sports hernias, examining both conservative and surgical approaches, as well as the role of rehabilitation in optimizing patient outcomes. By synthesizing the most recent evidence on sports hernias, this review aims to equip clinicians with the knowledge and tools necessary to identify, diagnose, and manage these challenging injuries effectively, ultimately improving patient care and facilitating a timely return to sport.

### *The impact of sleep interventions on athletic performance: A systematic review. Sport Medicine – Open, 9:58, 2023.*

**Background:** Sleep is essential for maximal performance in the athletic population. Despite that, the sport context has many factors that can negatively influence athletes' sleep and subsequent recovery.

**Objectives:** The purpose of this systematic review was to synthesize the most recent literature regarding sleep interventions aimed at improving sleep and subsequent performance in athletes.

**Methods:** The present systematic review was conducted based on the PRISMA guidelines and the PICOS approach. The search was conducted in May 2022 using the electronic database PubMed, SPORTDiscus via EBSCOhost, and Web of Science. Once extracted, studies were included if they met the following criteria: (1) participants were athletes of individual or team sports; (2) implemented an intervention

aimed at improving sleep; (3) measured at least one objective performance/recovery outcome; and (4) reported the relationship between sleep and performance.

**Results:** The search returned 1584 records. Following the screening, a total of 25 studies met our inclusion criteria. All the included articles were intervention studies published between 2011 and 2021. The included studies implemented various sleep interventions, such as sleep hygiene, naps, sleep extension, light manipulation, cold water immersion, mindfulness, or a combination of two or more strategies. Sleep extension and naps were the most representative and most effective strategies to improve sleep and performance. Mindfulness and light manipulation demonstrated promising results, but more studies are needed to confirm these findings. Sleep hygiene, removing electronic devices at night, and cold-water immersion had no effects on sleep and subsequent performance/recovery, but these results are based on a few studies only.

**Conclusion:** While acknowledging the limited amount of high-quality evidence reviewed, it appears that increasing sleep duration at night or through napping was the most effective interventions to improve physical and/or cognitive performance.